WTI CRUDE OIL FUTURES

Predictive Analysis for Energy Trading Strategies

Instrument Mechanics · Market Participants · Geopolitical Event Study · Predictive Models

Primary Audience: Energy Trading Desks | Data: FRED API · CFTC COT | Analysis Date: March 6, 2026

ROADMAP

Four Parts, One Coherent Story

Who should be in the room: Energy Trading Desks (primary) · Portfolio Managers · Corporate Hedgers

01

Instrument Mechanics

How WTI futures work
and what each $1/bbl move means
in dollars

02

Market Participants

How hedgers, speculators, and banks
position — and why divergence matters

03

Geopolitical Event Study

5 crises, 3 energy eras:
does US dominance compress crisis spikes?

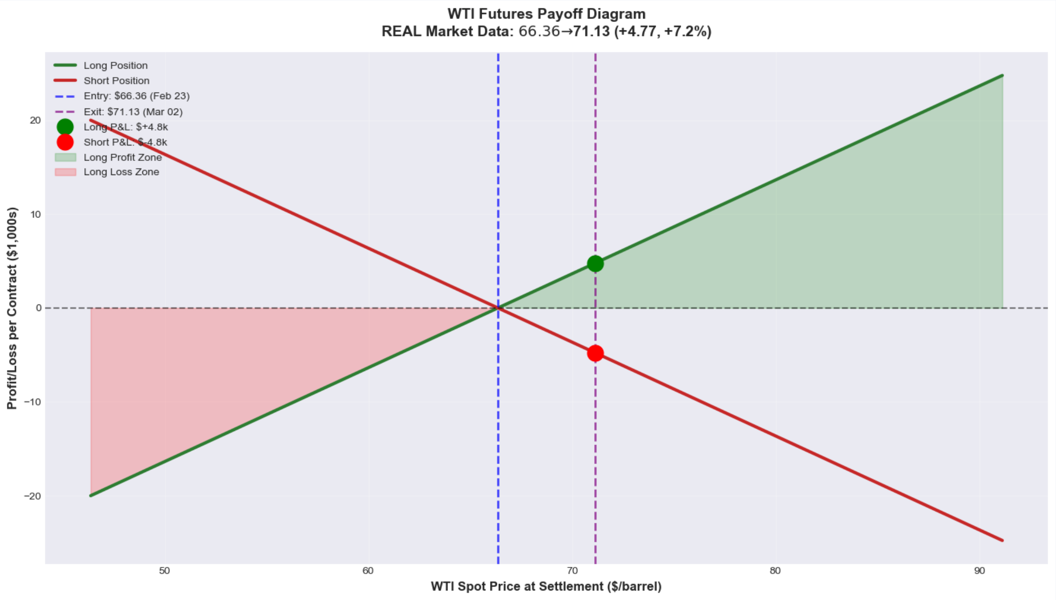
04

SQL Evidence + Models

12 SQL queries validate the narrative;
2 predictive models add short-horizon signals

Research questions: (1) Can technical indicators predict short-horizon WTI prices? (2) Has US energy independence reduced geopolitical risk premium? (3) Do participant cohorts position differently around crises? (4) Do momentum and volatility signals add predictive value?

One P&L Rule: Every $1/bbl Move = $1,000 Per Contract



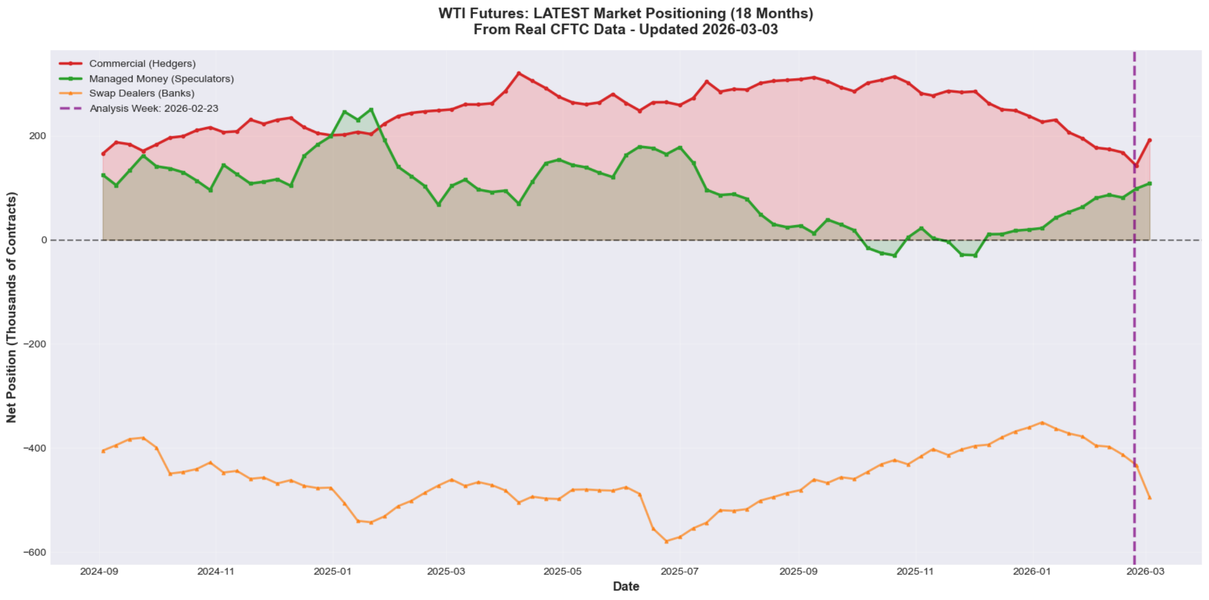
Contract Specs

Ticker	**CL (NYMEX)**
Size	**1,000 bbl/contract**
Delivery	**Cushing, Oklahoma**
Tick	**$0.01/bbl = $10/contract**
Last week	**$66.36 → $71.13 (+$4.77)**
Long P&L	**+$4,800 per contract**

So what?

Zero-sum structure means position sizing starts here: contracts = risk budget ÷ $1,000-per-$1 move.
Hedge ratio, stop distance, and take-profit levels all flow from this one translation.

Extreme Speculator-Hedger Divergence Signals Crowding — and Potential Reversal



Latest Week P&L
(real positioning × real price move)

Commercials
Long 168k net
+$801M

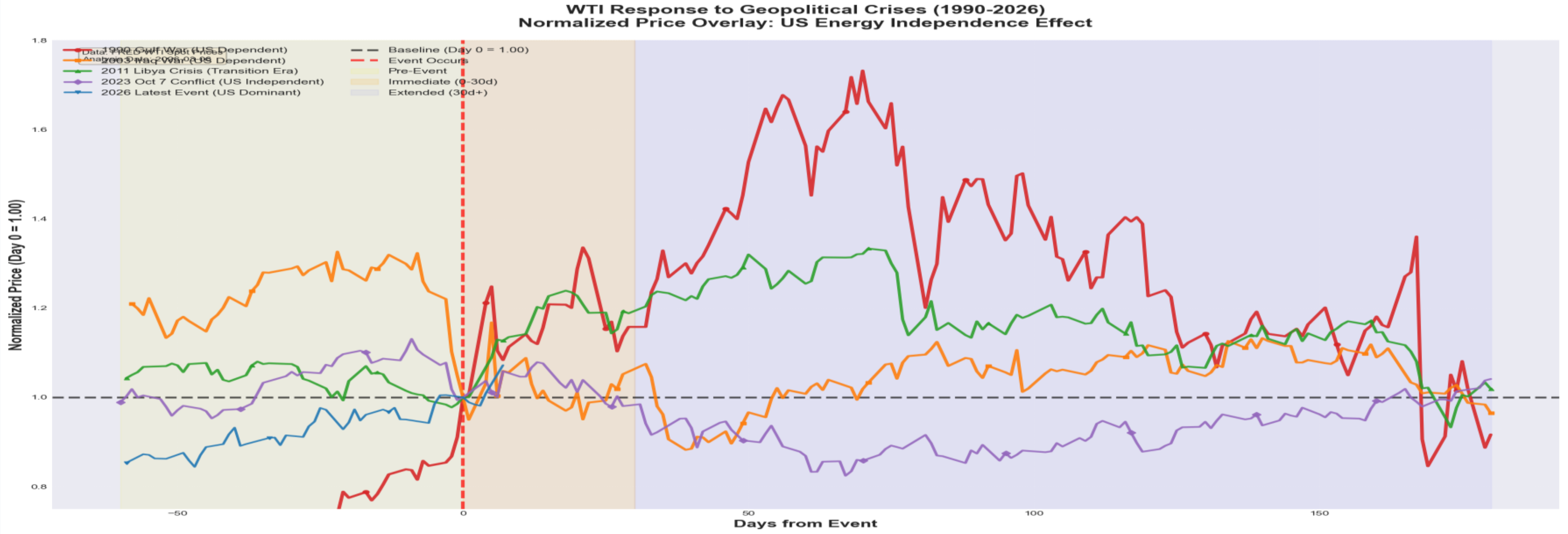
Managed Money
Long 81k net
+$387M

Swap Dealers
Short 413k net
−$1.97B

So what?

As of Mar 3 2026: speculators added length (108k net) while commercials also expanded (192k net). When both sides move together, crowding risk is lower — but Swap Dealers' record short (-543k) warrants monitoring. Treat extreme divergence as a regime flag before sizing.

US Energy Dominance Has Structurally Compressed the Geopolitical Risk Premium



US DEPENDENT (1990–2004)

Spikes: 17–34%

TRANSITION (2005–2013)

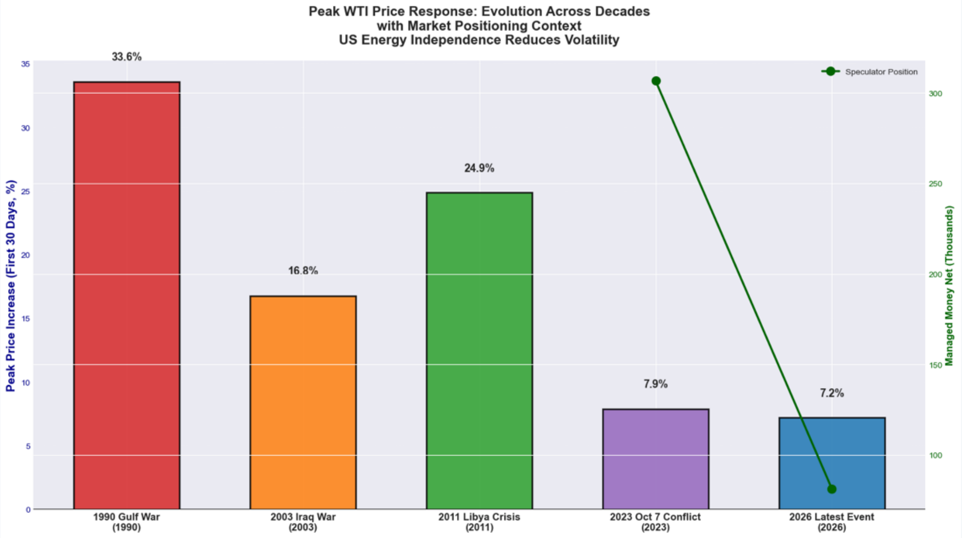
Spikes: 25%

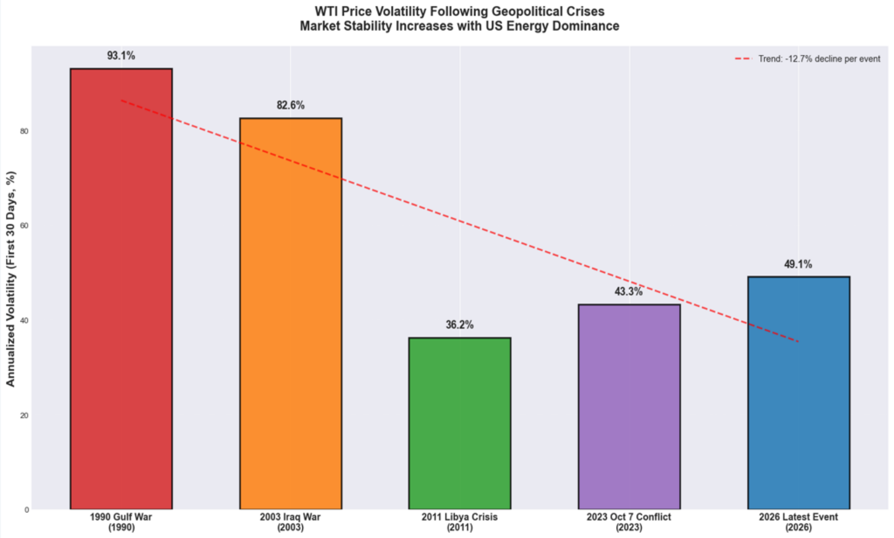
US DOMINANT (2019–present)

Spikes: 7–8%

Normalized to Day 0 = 1.00 at each event date. Older crises show much larger and more persistent spikes than recent events.

Peak Response Down 5× and Volatility Down 47% Since the US Dependent Era





34% → 7%

Peak 30-day response
1990 vs 2023/2026

93% → 49%

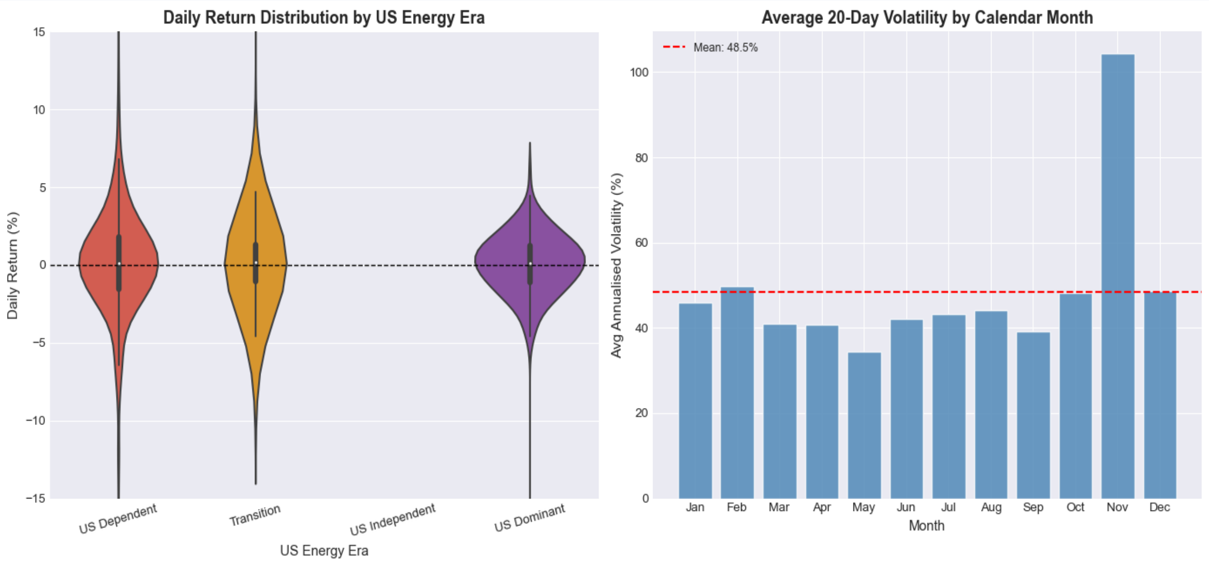
Annualised volatility
post-event window

No CFTC → 306k net long

Speculator position
at Oct 7 (crowded)

Implication: apply era-conditioned stress scenarios — 1990-style VaR shocks are no longer appropriate for geopolitical risk in the US Dominant era.

SQL and EDA Confirm: Era and Month Are the Two Key Regime Dimensions



SQLite Analysis
12 Queries Across 3 Tables

GROUP BY	Annual & monthly seasonality
Window fns	30-day rolling avg, price rank
JOINs	Event metrics + market context
Subqueries	Extreme-vol filtering by era

Key SQL Findings

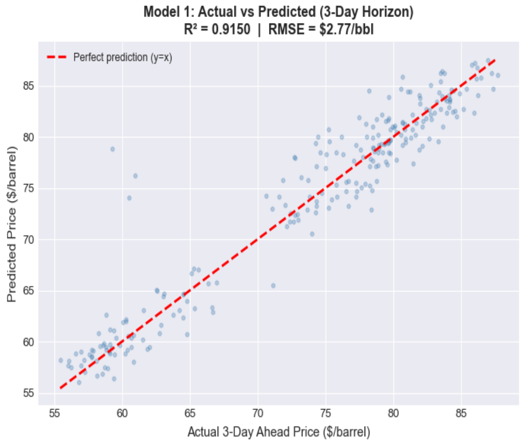
US Dominant era: 56% of days in Normal vol regime vs 31% for US Dependent — structurally calmer

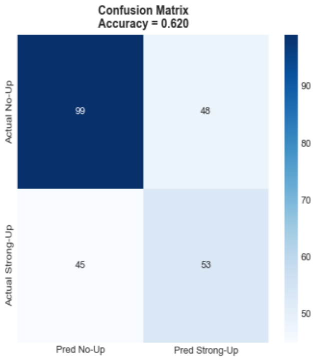
Near-event days (+30d): avg daily return 0.31% vs 0.19% on normal days — mild event premium persists

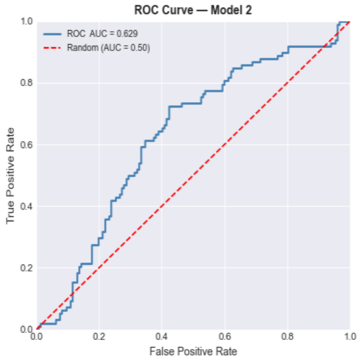
Rolling deviation from 30d avg: current WTI sits +7.4 above its 30-day average — elevated but not extreme

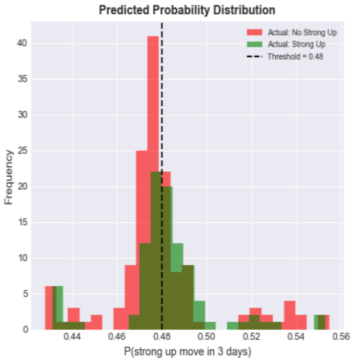
November historically 104% avg volatility — highest seasonal risk month; position size accordingly

Technical Signals Add Modest but Real Predictive Value Beyond Pure Autocorrelation









Model 1: Linear Regression

Predict WTI price level 3 trading days ahead

80/20 temporal split — no look-ahead

R^2 — **0.915**
91.5% of price variance explained

MAE — **$1.83/bbl**
vs. naive baseline of $8.26

RMSE — **$2.77/bbl**
≈ 4% of current WTI level

Model 2: Logistic Regression

Predict 3-day strong up-move (>+0.8%)

Threshold = 0.48; avoids look-ahead bias

ROC-AUC — **0.629**
vs 0.50 random baseline

Accuracy — **62.0%**
vs 60.0% naive baseline

Precision — **52.5% | Recall 54.1%**

THREE TAKEAWAYS + NEXT STEPS

What Every WTI Practitioner Should Act On

1 Re-calibrate geopolitical stress scenarios to the US Dominant era

Peak crisis response has fallen from ~34% (1990s) to ~7–8% (2023–2026). Applying 1990-era VaR shocks to today's book overstates geopolitical risk premium — freeing capital for other uses.

2 Use CFTC positioning divergence as a regime flag, not a standalone signal

Speculator-hedger divergence identifies crowding risk. Combine with event timing and volatility regime before adjusting position size or hedge urgency.

3 Layer technical signals into sizing decisions — not direction calls alone

Model 1 (R^2 = 0.92, MAE = $1.83) anchors a 3-day price range for entries. Model 2 (AUC = 0.63) tiers directional conviction. Neither is a standalone buy/sell trigger.

If more time: EIA crude inventories · WTI term-structure spread (roll yield) · OVX implied volatility vs realized — to improve out-of-sample model calibration.